



Bird poop and dust ruin Mars plans

SpaceX plans to use a heat shield that will leak liquid through small pores which could easily be clogged by dust. <https://digit.in/mar19-13>



Experimental battery on Moon

Japanese startup plans to test solid-state battery on the Moon's surface <https://digit.in/mar19-14>

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he square kilometer array (SKA) is a radio telescope being built for the purpose of understanding the origin and evolution of the universe. With an effective collecting area of one square kilometer, it is going to be the largest telescope ever built. The telescope is currently in the design phase, and the construction is expected to start in 2024. There are two phases of the construction, with the first phase designed for ten percent of the total capacity of the telescope. The costs of the second phase has not yet been worked out. The telescope is expected to start its observation runs in 2028. In South Africa, the MeerKAT telescope with 64 antennas is undergoing verification tests, and is expected to be the largest radio telescope in the

southern hemisphere, till the SKA is realised. The MeerKAT array will be absorbed into the SKA, and become a part of it. MeerKAT is just one of the many telescopes around the world, that are being built just so that scientists can get the experience necessary to design and build the SKA. Around the world, are several officially designated SKA pathfinder telescopes, that are engaged in observational campaigns and science activities similar to what the SKA is being designed for. These telescopes provide the feedback necessary to the design teams. One of these telescopes, the Giant Metrewave Radio Telescope (GMRT), is located in India.

INTERNATIONAL COLLABORATION

The building of the SKA is a global effort, that is already decades old. In 1993, the International Union of Radio Science (URSI) established the Large Telescope Working Group

to work out the details of building the next generation radio telescope. Between 1997 and 2007, a number of international collaboration agreements were drawn up towards the effort. The effort is headed by the SKA organisation, a non-profit company based in the UK. The twelve participating countries which are a part of the SKA organisation are Australia, Canada, China, France, India, Italy, New Zealand, South Africa, Spain, Sweden, Netherlands and United Kingdom. Academic institutions and government organisations from these countries are involved in the project. The National Centre for Radio Astrophysics is the participating institution from India.

The antennas of the SKA will be physically located in Australia, Botswana, Ghana, Kenya, Madagascar, Mauritius, Mozambique, Namibia, South Africa and Zambia. Apart from these, twenty other countries are participating in the technical aspects of the design and development of the telescope. These countries include Germany, Portugal and Japan. In all, representatives of an estimated 40 percent of the population of the world, is involved in some way or the other, towards the building of the SKA. It's one of the largest scientific endeavors ever undertaken.

SKA LAYOUT

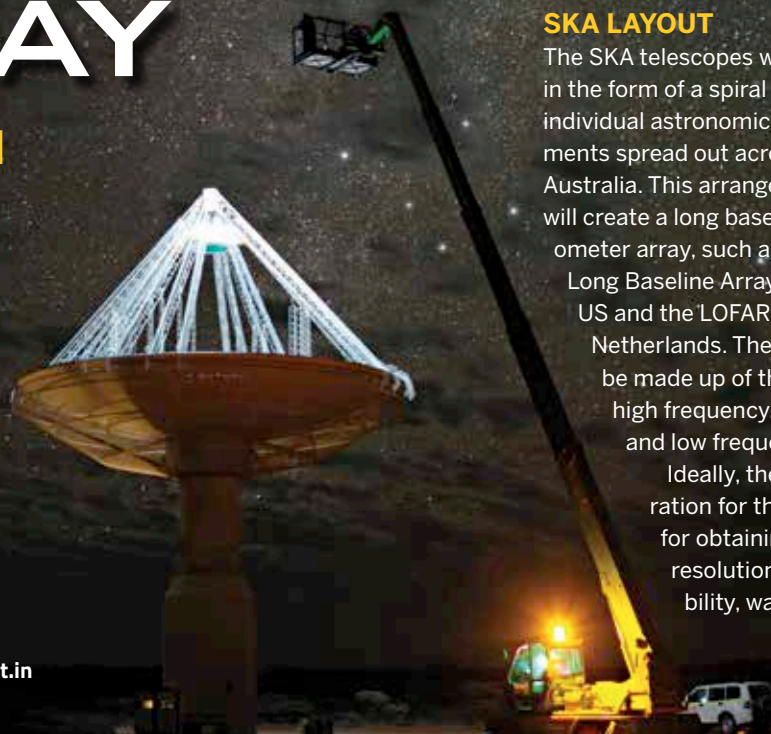
The SKA telescopes will be arrayed in the form of a spiral galaxy, with individual astronomical instruments spread out across Africa and Australia. This arrangement of dishes will create a long baseline interferometer array, such as the Very Long Baseline Array (VLBA) in the US and the LOFAR telescope in Netherlands. The telescope will be made up of thousands of high frequency, mid frequency and low frequency dishes.

Ideally, the best configuration for the telescope for obtaining the highest resolution imaging capability, was to randomly

THE SQUARE KILOMETER ARRAY

A continent sized radio telescope

Aditya Madanapalle | aditya@digit.in



Credit: SKA organisation